PhenoV Pro v1.0

A Comprehensive Framework for Automated Analysis of Plant Phenotypic Traits from Image

User’s Guide

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1 Overview of PhenoV Pro

PhenoV Pro is a plant image processing and analysis tool developed based on the Qt framework. It is designed to provide efficient and accurate analysis of specific plant categories and image formats through various computer vision methods. The software is suitable for research applications in fields such as agriculture and botany.

System Requirements

Operating System: Windows 7 or above.

Processor: Intel Core i5 or above.

Memory: 4GB or above.

Disk Space: At least 300MB of available space.

Display: Resolution of 1024×768 or above.

Software Startup

After downloading, double-click the software to open it. No login is required. The software will display the main page in a maximized window upon opening.

User Interface Overview

The main interface consists of:

Menu Bar: Located at the top, containing "Start", "Edit", "Tool", and "Help" menus.

Tab Bar: Displays currently open and active tabs.

Tab Area: Displays the functional interface of the selected tab (defaults to "Main").

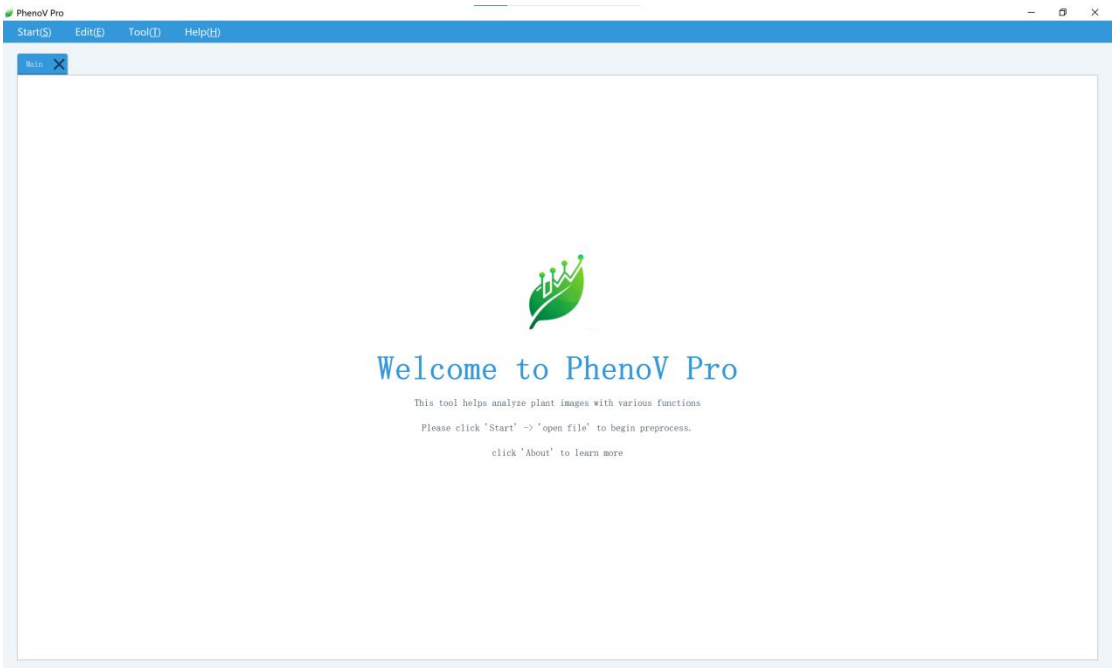

Fig.1 Main Interface

2 Image Preprocessing

The "Edit" menu (Alt+E) provides tools for initial image processing to meet analysis needs.

2.1 Grayscale

Function: Converts a color image into a grayscale image.

Shortcut: Ctrl+G.

Procedure:

1. Open an image via "Start -> Open".

2. Select "Edit -> Grayscale".

3. Enter the threshold parameter in the pop-up box.

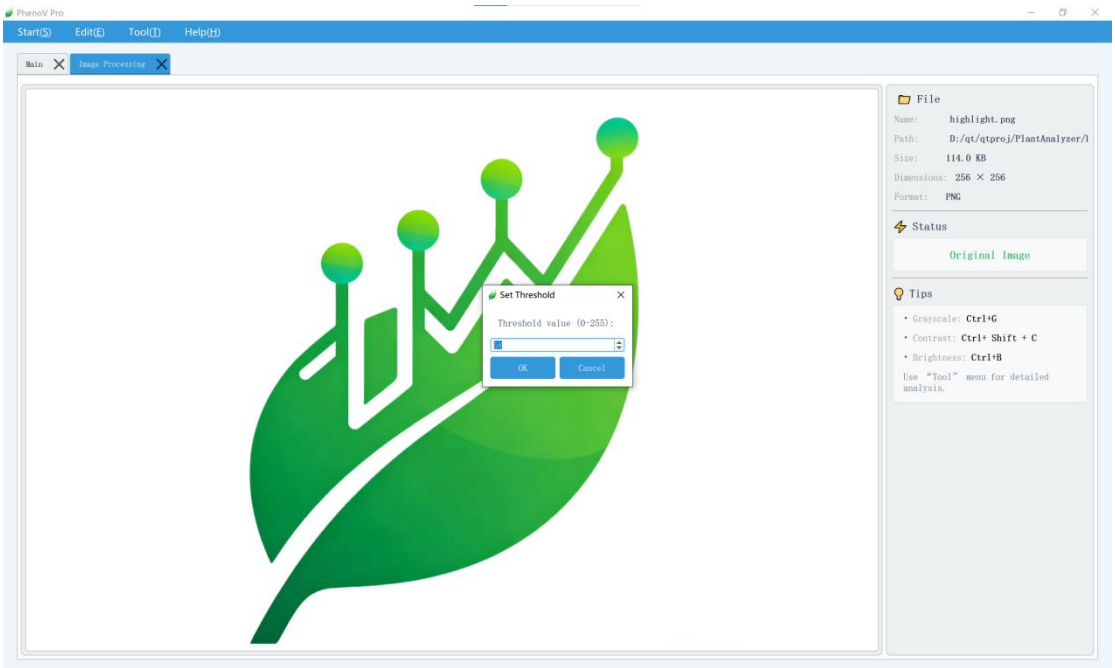

Fig.2 Setting Grayscale Threshold

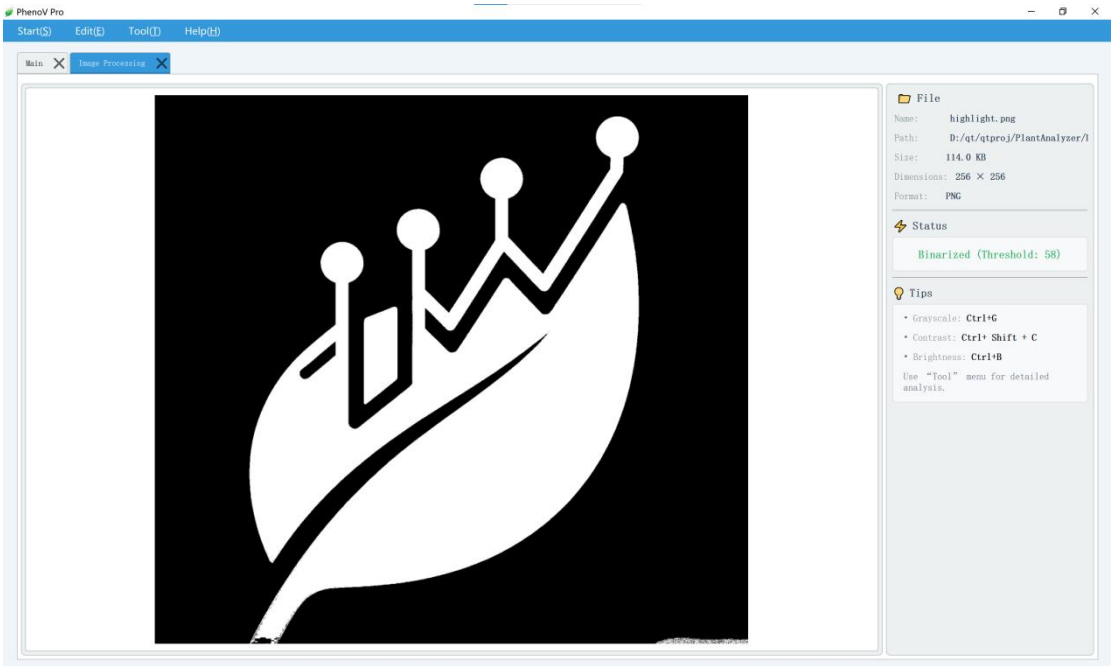

Fig.3 Grayscale Effect Display

2.2 Contrast

Function: Adjusts the image contrast.

Shortcut: Ctrl+C.

Procedure:

Follow the same steps as Grayscale, but select "Edit -> Contrast" and enter a contrast factor (0.01 to 10.0).

Fig.4 Setting Contrast Threshold

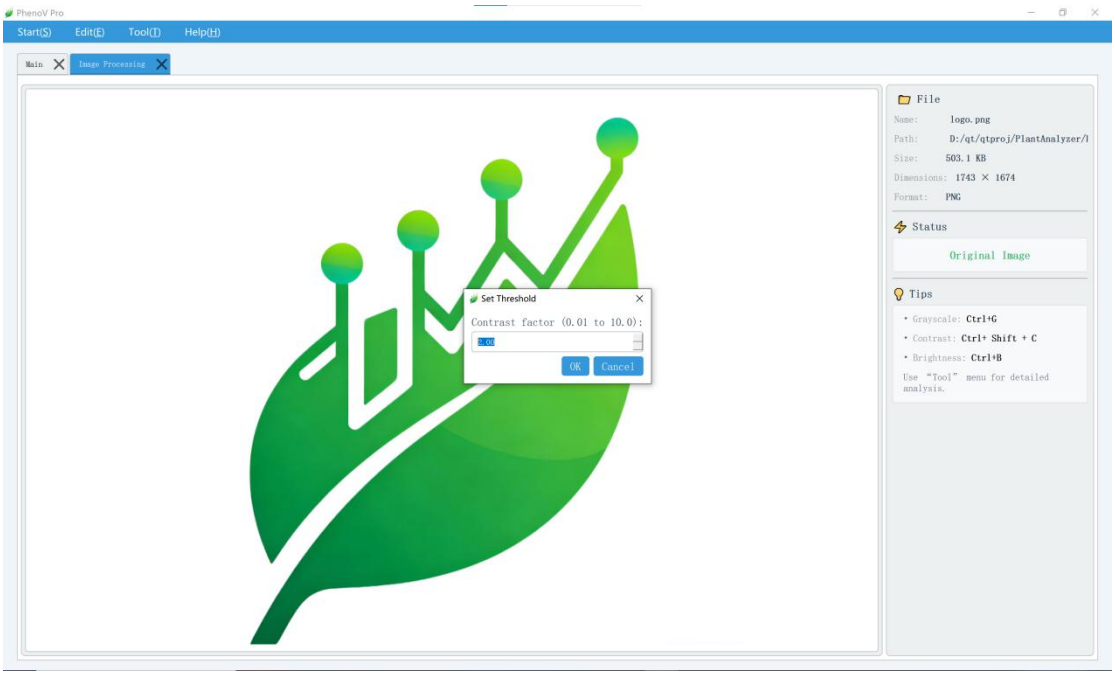

Fig.5 Contrast Effect Chart

2.3 Brightness

Function: Adjusts the image brightness.

Shortcut: Ctrl+B.

Procedure:

Follow the same steps as Grayscale,but select "Edit -> Brightness" and enter a brightness offset (-200 to 200).

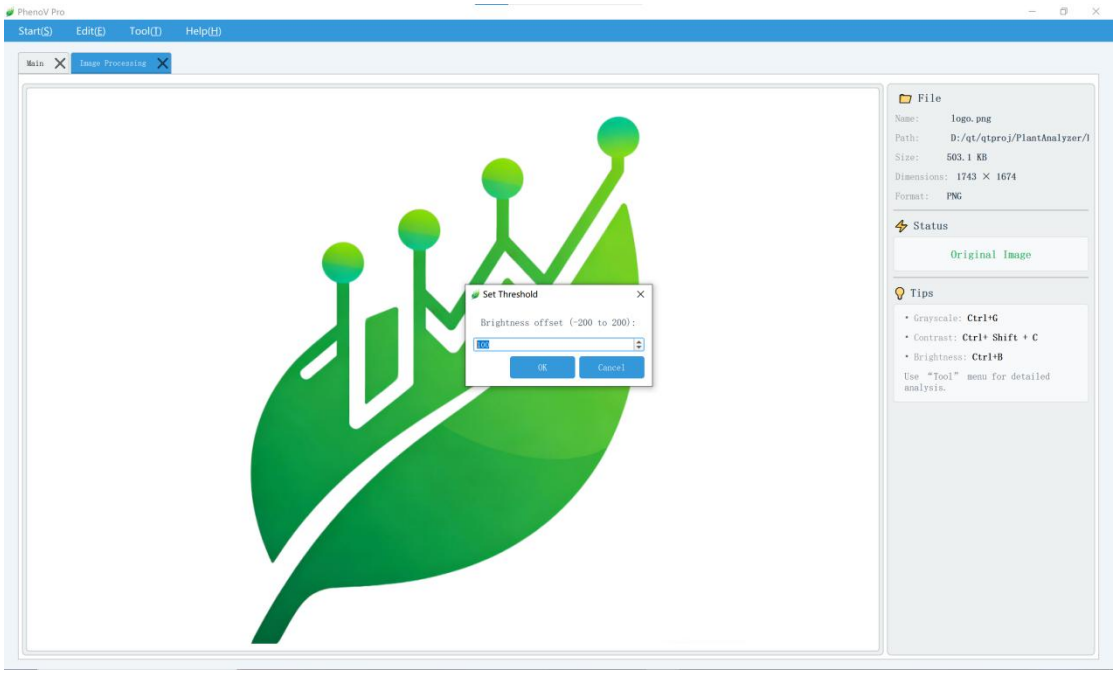

Fig.6 Setting Brightness Threshold

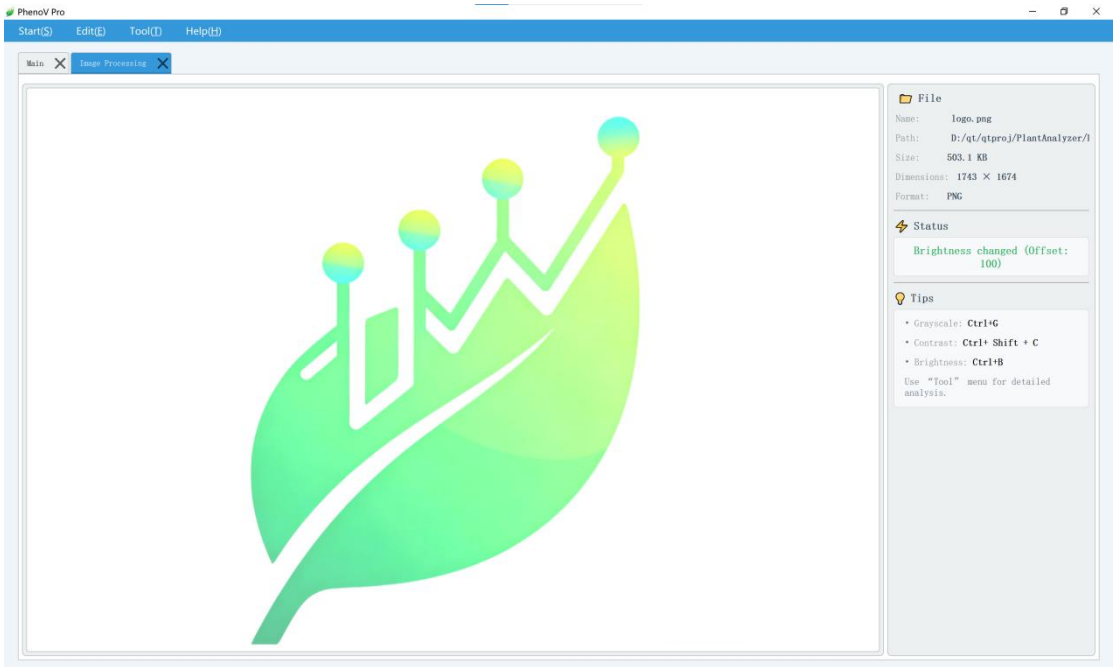

Fig.7 Brightness Adjustment Effect Chart

2.4 Target Isolation

Color Range

Function: Extracts specific image regions based on a selected color range. This is primarily used for segmenting plants (e.g., green leaves or roots) from the background.

Shortcut: Ctrl+Shift+C.

Procedure:

1. Follow the same steps as Grayscale, select "**Target Isolation-> Color Range**".

2. Click "OK" and click on the image to pick a target color.

3. Adjust the "Tolerance" slider to expand or shrink the selection.

4. click "Apply" to generate a binary mask.

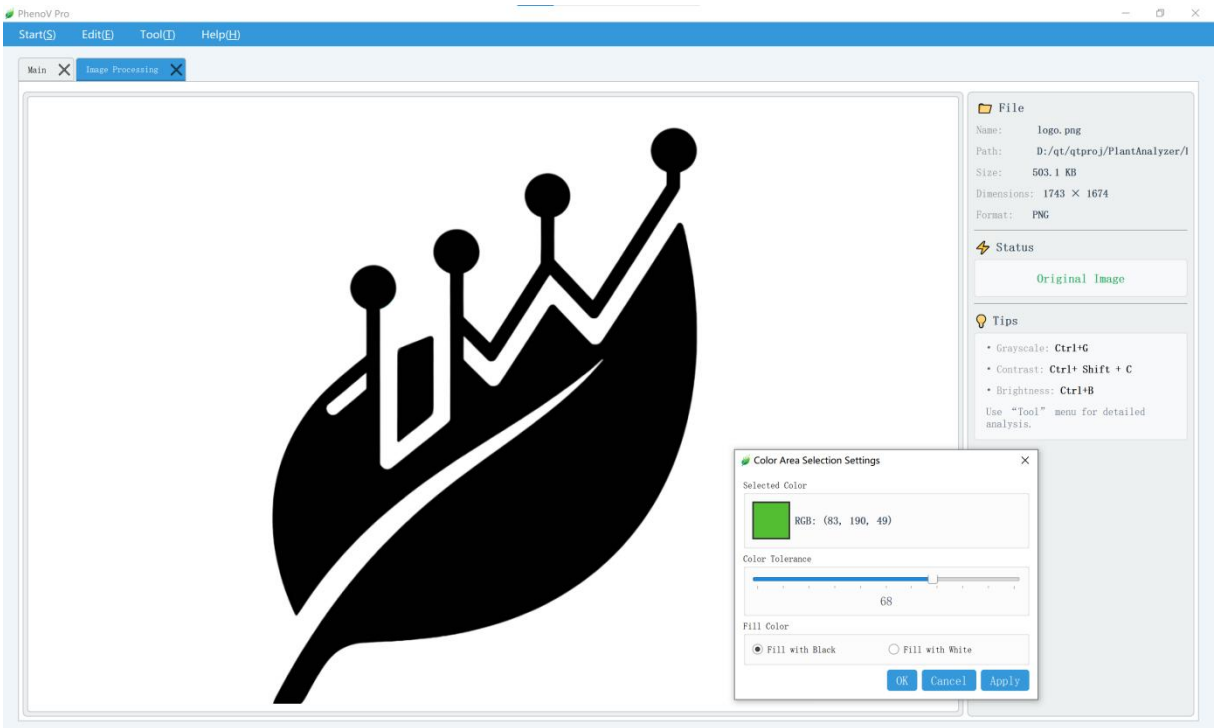

Fig.8 Color Range Effect Chart

Refine Brush

Function: Manually fine-tunes the segmentation results by painting or erasing specific areas. It helps to reconnect broken roots or remove background noise.

Shortcut: Ctrl+Shift+B.

Procedure:

1. Select "**Target Isolation-> Refine Brush**".

2. Choose the brush color in the dialog.

3. Adjust the "Brush Size", and drag the cursor over the image to modify the pixels.

4. Click "Apply" to save the changes.

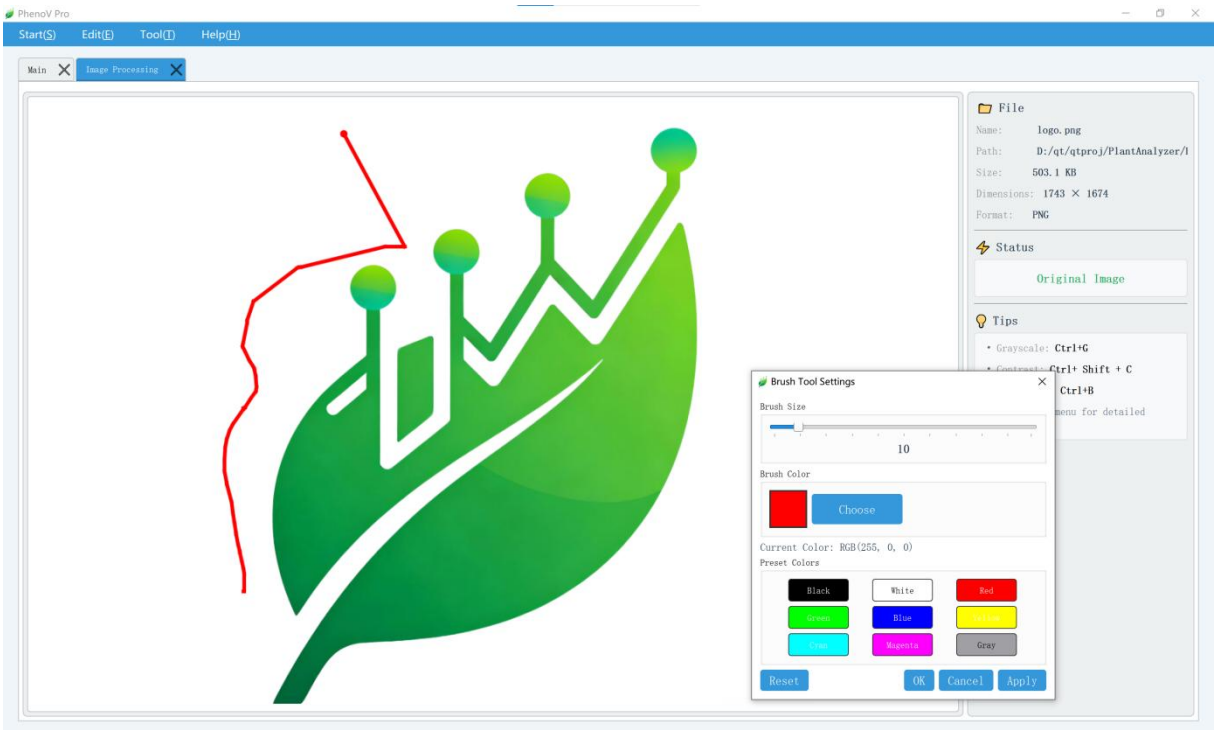

Fig.9 Refine Brush Effect Chart

3 Plant phenomics analysis

3.1 Seed Counting

Function: Automatically identifies and counts seeds in an image.

Shortcut: Ctrl+1.

Procedure:

1. Select "Tool -> Seed Count".

2. Open the image, and click "Analyze".

3. The results and contours will be displayed.

Fig.10 Seed Counting

3.2 Lesion Counting

Function: Identifies and counts lesions (disease spots) on plant leaves.

Shortcut: Ctrl+2.

Procedure:

1. Select "Tool -> Leaf Lesion Count".

2. Open the image, and click "Analyze".

3. Lesion contours will be identified in red.

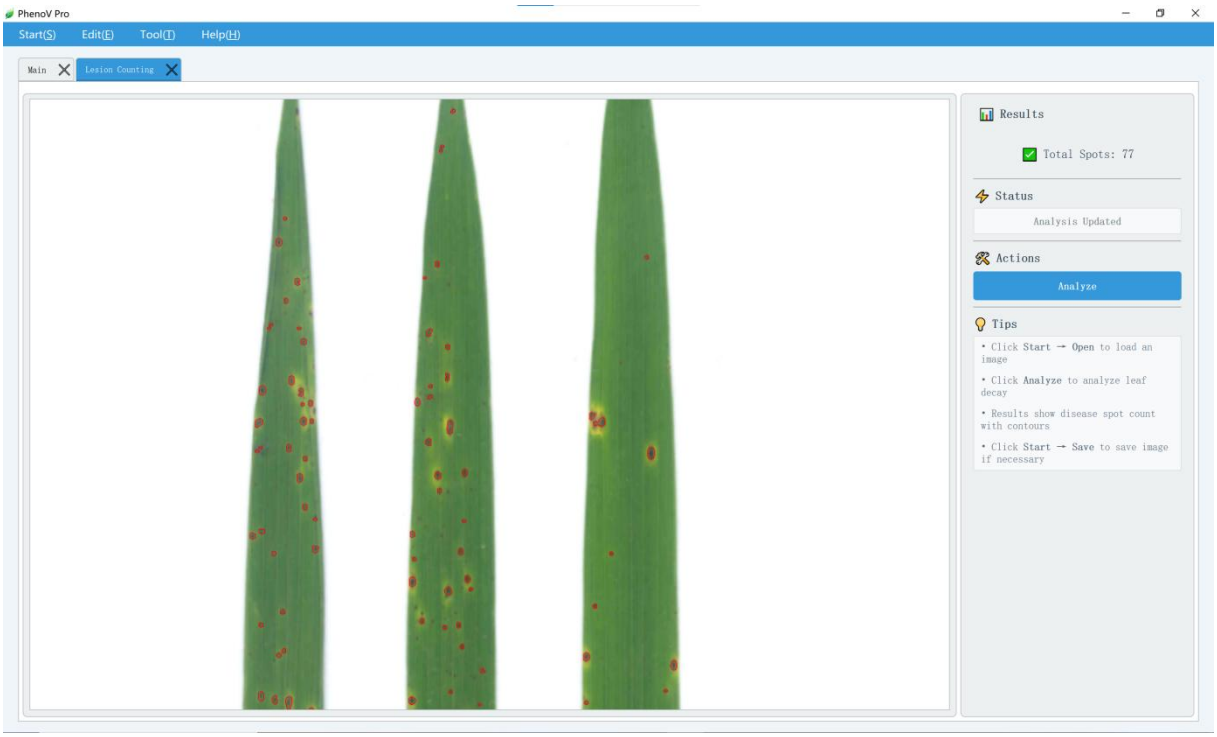

Fig.11 Lesion Counting

3.3 Lesion Area Measurement

Function: Analyzes the percentage of lesion area on a leaf.

Shortcut: Ctrl+3.

Procedure:

1. Select "Tool -> Leaf Area Computing".

2. Open the image, and click "Analyze".Leaves are outlined in orange and lesions in red.

3. Users can drag a slider to adjust the threshold for optimal identification.

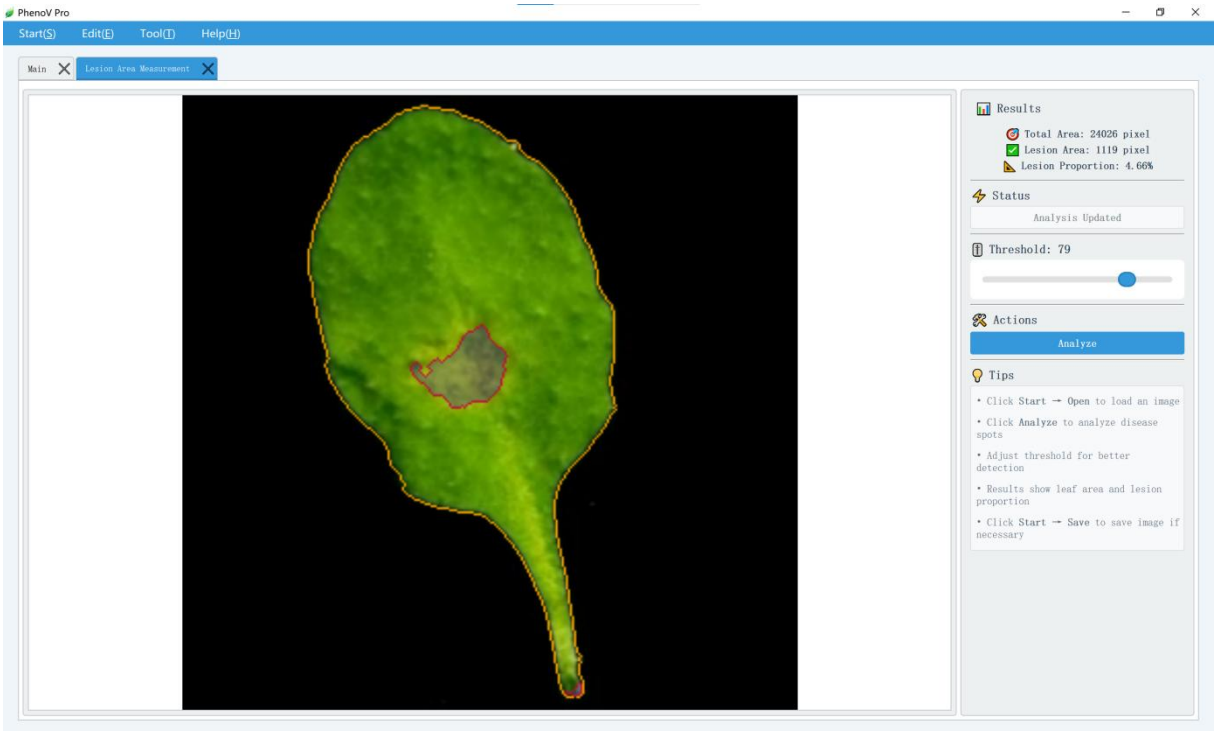

Fig.12 Leaf Disease Area Calculation

3.4 Leaf Growth Estimation

Function: Analyzes growth area changes by comparing control and target leaves.

Shortcut: Ctrl+4.

Procedure:

1. Select "Tool -> Leaf Growth Estimation".

2. Open the image, and click "Analyze".

3. Click a leaf to set as "Control" and enter its actual area.

4. Select "Target" leaves. A table will generate records with area and fold-change values.

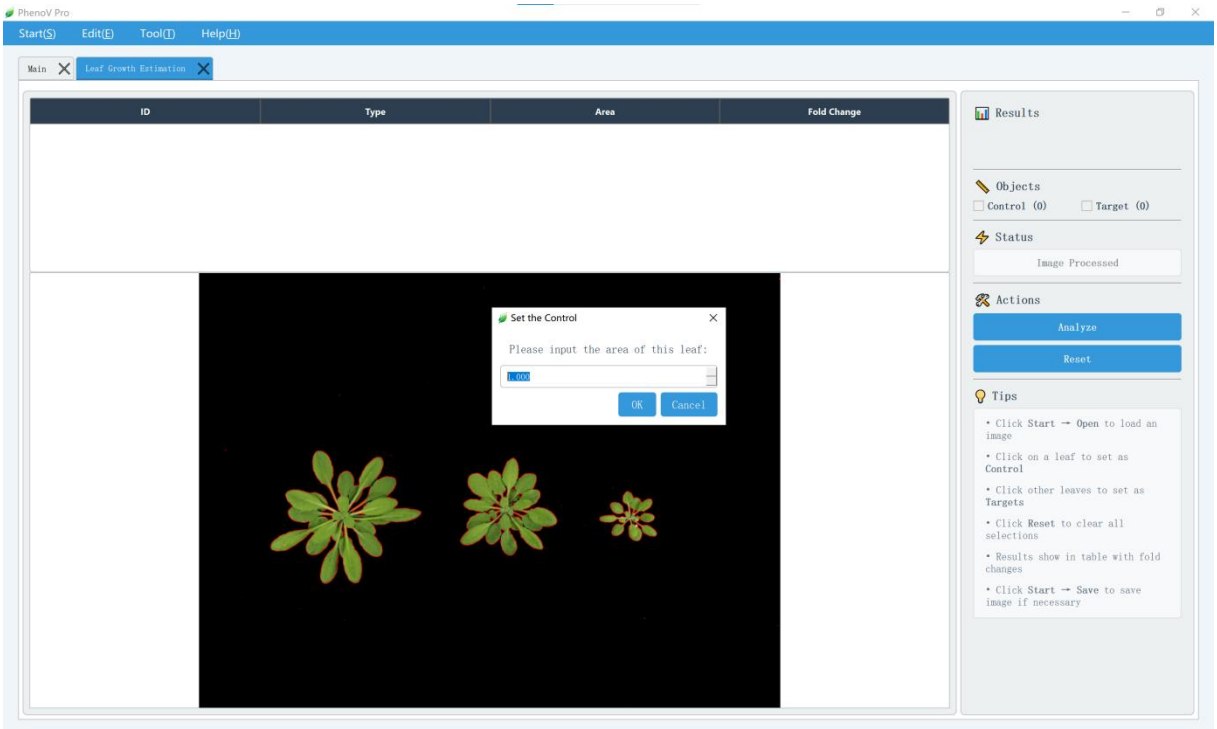

Fig.13 Setting Control Actual Area

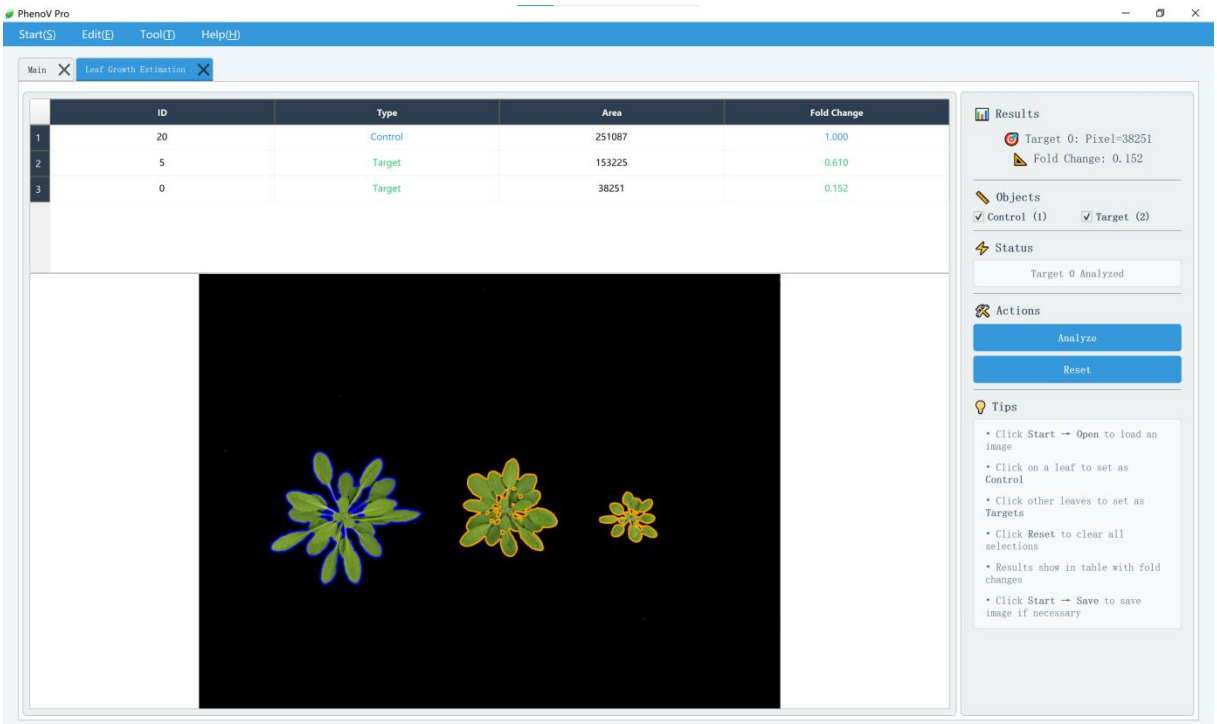

Fig.14 Leaf Area Conversion Table

3.5 Root Analysis

Function: Identifies plant roots and calculates their length.

Shortcut: Ctrl+5.

Procedure(Traditional mode):

1. Select "Tool -> Root Analysis".

2. Open the image, and click "Process" to see a grayscale mask.

3. Click on the plant to identify the contour and Main Branch.

4. Click root segmentation points to complete the calculation.

In the **Deep Learning mode**, just click "process"after open the image.

Fig.15 Grayscale Mask

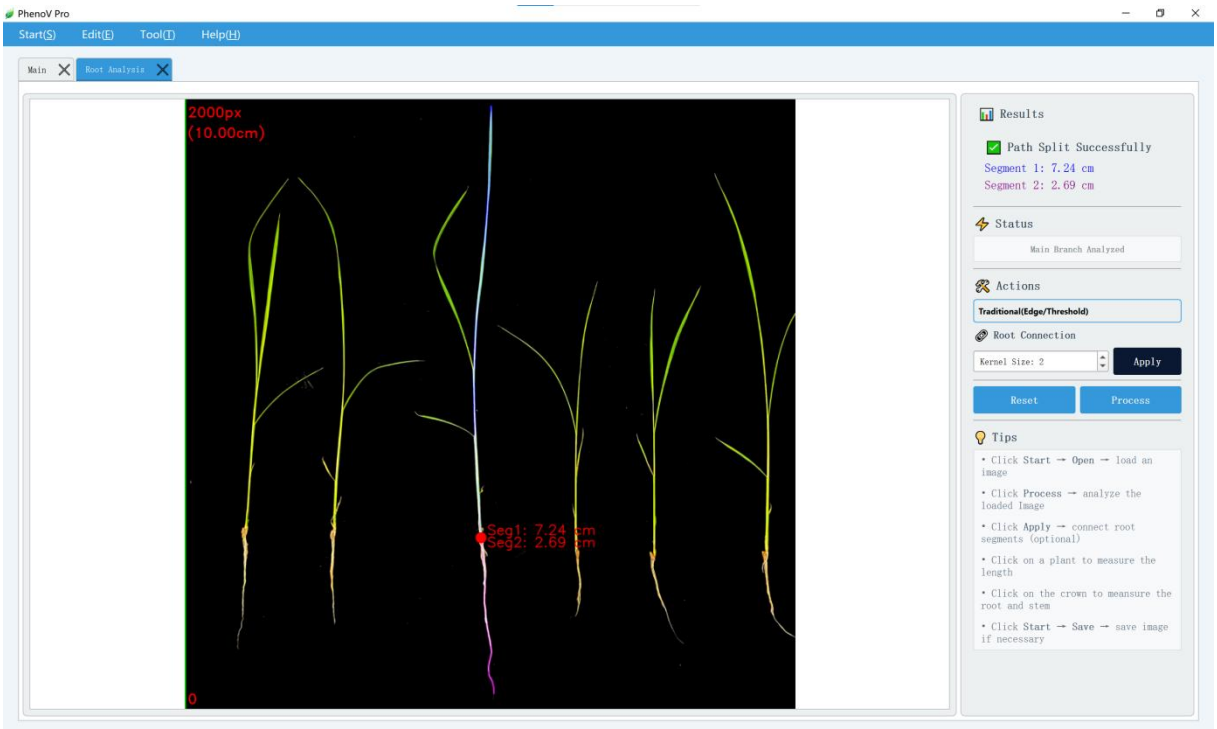

Fig.16 Root Length Segmentation and Calculation (Traditional mode)

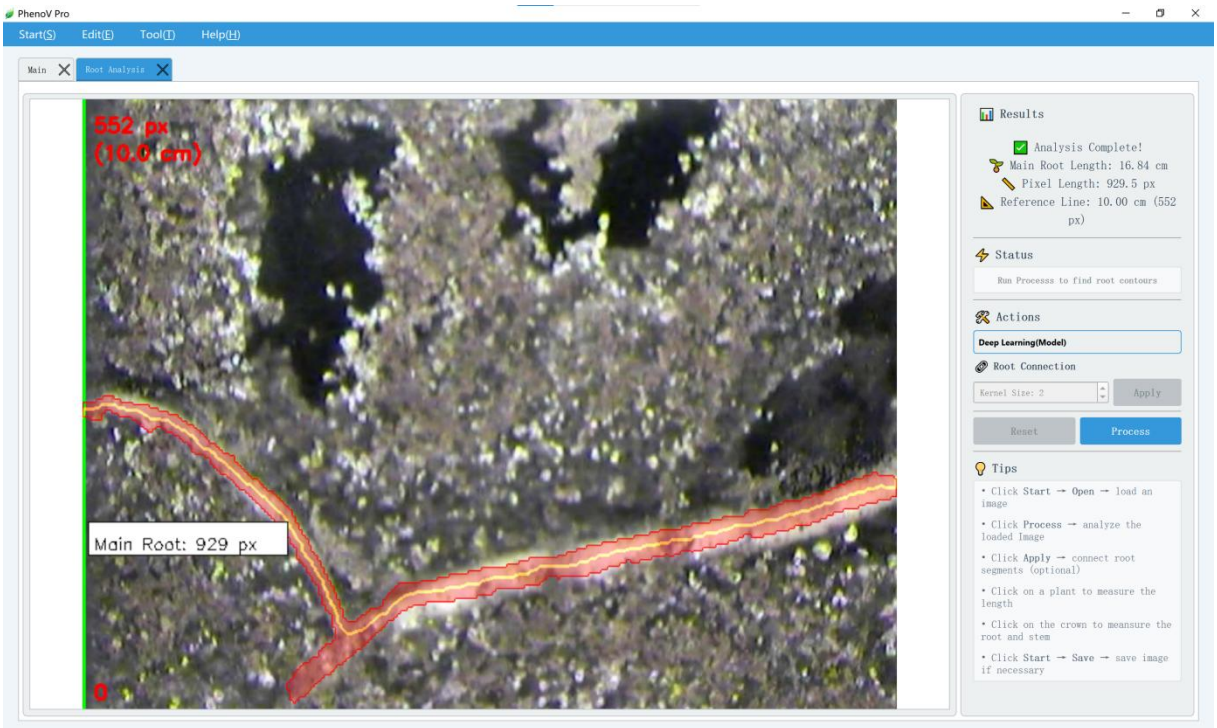

Fig.17 Root Length Segmentation and Calculation (Deep Learning mode)

4 Help and Support

The **Help (Alt+H)** menu includes "Help" and "About" options to view operating instructions and development team information. To install the software on a computer, follow the installation wizard prompts and click "Next" to complete the setup.